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Research Proposal on A Morphosyntactic Filter for Santali: Prompt-Driven Large Language Model Parsing for Universal Dependency Treebanks

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Abstract

While Large Language Models excel in better-resourced languages, they often exhibit significant typological deterioration in low-resourced languages. Due to the inadequacy of digital data, these models often hallucinate the structures of dominant languages onto low-resourced, morphologically rich languages like Santali, effectively erasing the key Austro-Asiatic morphological distinctions. This research proposes a morphosyntactic prompt filter constructed to constrain the Large Language Model generations within language-specific parameters before analysing Universal Dependency parses. The study adopts a hybrid framework combining usage-based morphology and functional-typological grammar. Usage-based approaches treat languages as emergent, frequency-driven schemas rather than mere static rules. Functional-typological grammar explains Santali's agglutinative morphology in terms of discourse functions and argument-structure mapping. Methodologically, this research involves constructing a corpus of 500-1000 sentences, annotated via ELAN and FLE_x, and transforming it into Universal Dependency format. Prompt templates are then designed using Chain of Thought (CoT) reasoning that forces the model to recognise Santali-specific morphosyntactic structures before parsing. The evaluation is conducted on Large Language Models such as GPT-4o, Claude 3.5 Sonnet, and DeepSeek-V3 in baseline and filter-guided conditions to see the output after prompting. Parsing performance is assessed using UAS/LAS and qualitative error analysis, drawing on cross-linguistic projection. The project illustrates how linguistically encoded prompting can mitigate hallucinations, support Universal Dependency-style parsing of Santali, and prevent the digital eradication of an agglutinative Austro-Asiatic language.

Keywords: Large Language Models, Santali Morphosyntax, Prompt Filter, Digital Language Documentation, Low-Resource Language, Cross-Language Projection.

1. Introduction

Large Language Models illustrate near-human performance in high-resource languages, for instance, English, Chinese, etc. Nevertheless, for low-resource languages (low-resource languages) like Santali, these models frequently fail to produce accurate output. Because of inadequacies in digital training data and literature, Large Language Models often hallucinate linguistic structures, imposing the grammar of dominant languages (like Bengali or English) on Santali. This project proposes a linguistic solution to a computational problem: developing a grammatical prompt filter that constrains the Large Language Model to recognise the explicit morphosyntactic rules of Santali. It will also help the Large Language Model to parse Santali outputs accurately, understand the speaker's linguistic intent and respect the Santali-specific markers, aspects and rules before generating analyses and Universal Dependency parses. By integrating both Descriptive Linguistics and Computational Interaction Theory, this research will leverage the digital archiving of Austro-Asiatic languages through prompt-based filters to reduce linguistic erasure.

2. Statement of the Problem

When translating or analysing Santali, recent Large Language Models often fail to correctly identify complex verbal morphology (e.g., subject/object agreement markers or the topic marker).

A Large Language Model might translate a Santali sentence by guessing meanings based on Bengali or English cognates, neglecting the exclusive affixation that determines the actual tense and aspect. This results in linguistic erasure, where the marked grammatical identity of the indigenous language is absent in digital translation. Santali is extremely underrepresented in digital databases, especially compared to Bangla or English. Due to this underrepresentation, the Large Language Model models frequently hallucinate the structure of Santali syntax based on high-resource languages. The Large Language Models assume the pattern well presented in their training data, implying the morphosyntax of better-resourced languages in the analysis of under-resourced ones. This issue indicates that the models view the sentences but interpret the structures based on Bangla or English grammar rather than the genuine Santali systems.

3. Research Question

3.1 Primary Research Question: How can a prompt-driven linguistic filter enable Large Language Models to parse a low-resource language like Santali and generate Universal Dependency treebanks with fewer errors?

3.2 Secondary Research Questions:

3.2.1 How effectively do Large Language Models identify marked features and complex structures of Santali via morphosyntactic prompts?

3.2.2 What specific morphosyntactic prompt templates should be provided to Large Language Models to prevent linguistic hallucinations and the projection of dominant language rules onto Santali?

4. Research Objectives

The ultimate goal of this study is to build an effective prompt filter for agglutinative languages, where Large Language Models often fail to recognise their linguistic features. The objective of this research is as follows:

4.1 To develop and validate a prompt-driven morphosyntactic filter for the Santali language that enables Large Language models to parse Santali sentences and generate Universal Dependency treebanks accurately.

4.2 To evaluate the ability of Large Language Models in identifying complex markers of Santali, such as topic markers, TAM, and object identification through linguistic prompts, thereby mitigating linguistic erasure or hallucination found in standard Large Language Model outputs.

4.3 To design and formulate a set of linguistically grounded prompt templates for Santali marked features and grammatical markers, and assess the effectiveness of Large Language Models when guided by these specially developed prompts.

4.4 To compare the parsing accuracy of the developed prompt-driven linguistic filter against baseline zero-shot Large Language Model outputs to assess the minimization in syntactic errors via metrics like UAS and LAS.

5. Literature review

Large Language Models show significant insufficiency in low-resource languages due to limited resources. Large Language Models are trained through the available resources found on the internet. Despite being one of the languages with a large number of speakers within the Munda subgroup of the Austro-Asiatic language family, Santali is predominantly neglected and unstudied in modern NLP research (Billah et al., 2024). They also stated that low-resource languages are primarily those that lack extensive monolingual or parallel corpora, in addition to the manually generated language resources required for creating statistical NLP applications. There are several corpora in Santali, yet it is recognised as a low-resource language. Moreover, these corpora somewhat lack the variation of Santali used in the Northern part of Bangladesh and are more India-centric.

Mujadia and Sharma (2024) developed multilingual, multitask translation models for 36 Indian languages, intermingling translation, grammar correction, error identification, quality estimation, and post-editing, ensuring robust cross-lingual communication where Santali was one of the targeted languages in the ecosystem. However, the work solely focused on machine translation rather than morphosyntactic prompting or Universal Dependency-style parsing. As per Zhuo et al. (2023), the performance of prompt-based semantic parsing with large pre-trained language models is sensitive to prompt design. Prompt-based learning is an alternative approach to supervised learning that aims to reduce the reliance on large human-annotated datasets (Liu et al., 2021). Unlike traditional supervised models, which estimate the probability of an output given an input text, prompt-based learning models estimate the probability of the text directly. This is achieved by applying prompt functions to modify the input text into various prompt templates with unfilled slots (Zhuo et al., 2023).

Ghosh (2008) identified complex morphosyntactic structures of Santali, such as case markers, definiteness, agreement markers, TAM, verb serialization, complex sentence structures, etc. Furthermore, Santali possesses direct objects marked by the clitic pronominals in infixed forms and indirect objects marked by an applicative -a. These distinctions are not recognised by Large Language Models due to limited resources. Though there is a growing body of research on training Large Language Models for low-resource languages and translating high-resource languages to low-resource languages, there are significant disparities in developing filters to provide accurate output for low-resource languages. However, there is almost no research on Santali morphosyntax using prompt-based filters, nor are there works examining how the prompt-based structures refine the Large Language Model-based parsing or Universal Dependency style analysis to prevent the linguistic projection from high-resource languages. In addition, Universal Dependency does not properly cover the unique features, such as the complex tri-number system and animacy/inanimacy in Austro-Asiatic languages. This research will explore how to

create a detailed filter for Santali morphosyntactic features using complex language prompts and help Large Language Models build Universal Dependency treebanks accurately.

6. Theoretical Framework

This study endorses a hybrid theoretical framework that includes usage-based morphology (Bybee, 1987) along with functional-typological grammar (Givón, 1984,1990; Greenberg, 1963) in order to model Santali morphosyntax and its interactions with Large Language Models. In usage-based models, morphology is a system of schemas derived from actual patterns of speech. Language is viewed as emerging from actual usage events where grammatical structures are modelled through frequency, entrenchment and distributional patterns in experience. Schemas become productive due to the reinforcement of massive numbers of existing examples. High-frequency words are stored as whole units, while low-frequency words are computed using patterns in the human brain. Santali suffixes are treated as discrete blocks in formal models. Usage based model is relevant to Santali, whose rich agglutinative morphology arises as recurrent patterns in speakers' usage. The morphosyntactic filter proposed in this study will operationalise this perspective by prioritizing high frequency constructional patterns instead of calculating every possible pattern from scratch, making them available to Large Language Models otherwise lacking sufficient Santali exposure.

Complementing this, functional typological grammar shapes the sentences by their communicative functions in which grammatical forms are linked to information structure, referential tracking and clause combining. Santali is a gold mine for functional typological grammar due to its head-final structure and unique features such as animate/inanimate distinctions and tri-number. When Large Language Models misinterpret the topic markers or affixations in Santali, they are not only marking formal errors but also misinterpreting the functional categories. The proposed filter foregrounds functionally loaded markers as requisite interpretive stages: prompts train the model to identify functional patterns, interpret their discourse and argument-structure functions and then assign Universal Dependency relations.

In accordance with Radical Construction Grammar (Croft, 2001), where categories are language-specific and construction-specific, this procedure treats Santali categories as construction-specific rather than as direct copies of other dominant language categories, and maps them onto Universal Dependency labels as per the Santali morphosyntactic system. These combined perspectives provide a fundamental boundary for designing morphosyntactic prompts and evaluating whether such linguistically encoded constraints can reduce hallucination and cross-linguistic projection in Large Language Model-based parsing of Santali.

7. Methodology

The research will follow a mixed-methods approach combining Descriptive Linguistics and Computational Prompt Engineering. The methodology is divided into three stages:

- Data Curation
- Filter Engineering

- Evaluation

6.1 Data Collection

- **Primary Data:** The research will integrate naturalistic speech of Santali used on a daily basis from the speakers in Northern Bangladesh (Gopalpur, Rajshahi).
- **Secondary Data:** Data will be extracted from established literature on Santali, such as Ghosh (2008) and the Santali Bible (available online).
- **Corpus Size:** A balanced set of data of 500-1000 sentences will be compiled through systematic collection of case markers, complex verb conjugation, TAM markers, aspect, etc. The corpus will consist of various morphosyntactic markers of Santali.

6.2 Annotation and the Development of the Gold Standard

The collection of data will be annotated manually to establish the ground truth against which the Large Language Model outputs will be measured.

6.2.1 Transcription and Segmentation: For the extracted data, ELAN will be used to segment the data into sentence-level units.

6.2.2 Interlinear Glossing: FLEEx will be used for interlinear glossing of the segmented texts.

6.2.3 Universal Dependency Conversion: The FLEEx output will be mapped to the Universal Dependencies (UD) format (CoNLL-U) for tagging relations like nsubj, obj and aux.

6.3 Designing the Filter (Prompt Engineering)

This phase translates the linguistic rules into computational prompts (the morphosyntactic filter).

6.3.1 Rule Extraction: Based on the FLEEx analysis, explicit constraints will be generated. For instance, if the negation *ba-ŋ* appears, tag it as NEG-1SG.

6.3.2 Template design: Prompt templates will be generated using Chain-of-Thought (CoT) reasoning. The prompts will instruct the Large Language Model to-

- **Analyse:** Identify the root verb and affixes first.
- **Constraint:** Apply the specific grammar rules extracted from Ghosh (2008).
- **Generate:** Output the final Universal Dependency parse tree after these checks.

6.4 Experimentation and Evaluation

6.4.1 Models: The experiment will be tested on prominent Large Language Models like GPT-4o, Claude 3.5 Sonnet, Deepseek-V3, etc.

6.4.2 Experimental Conditions:

- **Baseline:** The Large Language Model is asked to parse the Santali sentences without providing any linguistic rules.
- **Experimental:** The Large Language Model is asked to parse the Santali sentences using the morphosyntactic filter.

6.4.3 Metrics: The accuracy will be measured using standard parsing metrics-

- **UAS (Unlabeled Attachment Score):** Did the model link the correct words?
- **LAS (Labeled Attachment score):** Did the model assign the correct grammatical relation?
- **Error Analysis:** A qualitative review will quantify the occurrences where the model imposed Bangla or English grammar onto Santali syntax structure.

8. Ethical Considerations

The information of the participants will remain anonymous in the public dataset. Pseudonyms will be used to protect participants' identities. Under no circumstances will their personal information be revealed or leaked. The participation will be voluntary, and the participants will be free to withdraw at any time. All the transcripts and recordings will be kept safely with limited access. The secondary resources of data will be used, maintaining ethics. Furthermore, the research will adhere to the principles of data sovereignty, ensuring that the curated corpus will benefit the Santali community and ensure the prevention of linguistic erasure. The literature used during the procedures will be properly credited, assuring clarity and transparency.

9. Limitation of the Study

This present study, while contributing new insights to Large Language Model-based parsing in low-resource morphologically enriched language, possesses several limitations that should be acknowledged and considered when interpreting results:

- The corpus, consisting of 500-1000 sentences, may contain all the variations of Santali used in Northern Bangladesh and not all the regions where Santali is being used (Odisha, Jharkhand, etc.). Although the corpus size is a gold standard for training the Large Language Models, it may not contain all the linguistic variations.
- The evaluation is limited to some Large Language Models, and the results may not generalise to future models and open-source models, including Llama and Mistral. Findings are specific to Large Language Models evaluated in 2024–2025.
- Universal Dependencies, designed mostly for Indo-European languages, may not properly capture the Santali constructions.
- Only zero-shot and few-shot baselines will be tested; other baselines may not be explored due to time constraints.

10. Significance of the Research

This research shifts beyond the mere translation of high-resource languages to low-resource languages. The Austro-Asiatic languages resemble basic grammatical rules. This filter will help to build the

groundwork for other language families by applying the same methods employed in this research. Moreover, the Santali is one of the largest indigenous communities in Bangladesh, which is neglected compared to the CHT groups. This filter will provide a tested methodology and initial resources that can further support NLP research on Santali as well as other Low-Resource Austro-Asiatic languages in Bangladesh. It will also work for Sino-Tibetan or Indo-Aryan languages just by swapping the grammatical rules, subject to proper adaptation and validation. It verifies that we can build advanced AI tools accessible for indigenous language research without waiting for massive datasets to be built. The research demonstrates that, instead of traditional approaches that depend on massive parallel corpora, Linguistic Rule-Based Prompting can be used as an alternative to expensive model training. By creating a linguistically validated digital resource for Santali, the research addresses the active prevention of digital erasure and represents Northern Bangladesh-specific Santali in global AI databases.

11. Construction of the Dissertation

The structure of the study is outlined below:

Chapter 1: Introduction

- 1.1 Background of the study
- 1.2 Problem statement
- 1.3 Research questions and hypotheses
- 1.4 Objectives of the study
- 1.5 Rationale for a morphosyntactic prompt filter
- 1.6 Scope and delimitations
- 1.7 Organisation of the dissertation

Chapter 2: Literature Review

The existing literature will be reviewed systematically, which follows the framework of the study.

Chapter 3: Theoretical Framework

- 3.1 Santali morphosyntax
- 3.2 Morphological pattern of Austro-Asiatic languages and typological profile
- 3.3 Low-resource NLP and work on Santali and related languages
- 3.4 Large Language Models, hallucination, and cross-lingual transfer in low-resource languages
- 3.5 Prompt-based learning and semantic parsing with Large Language Models
- 3.6 Usage-based morphology (Bybee): patterns, frequency, entrenchment
- 3.7 Functional-typological grammar (Givón): form-function mappings
- 3.8 Constructional/typological insights (Croft): construction-specific categories
- 3.9 Synthesis: hybrid framework for a morphosyntactic filter

Chapter 4: Data, Field Site and Construction of the Corpus

- 4.1 Overall research design and paradigm
- 4.2 Field site: Northern Bangladesh Santali
- 4.3 Speech community, participants, and inclusion criteria
- 4.4 Primary data: naturalistic and elicited Santali speech
- 4.5 Secondary data: grammars, Bible text, existing corpora
- 4.6 Data collection procedures and recording setup
- 4.7 Transcription and segmentation with ELAN
- 4.8 Interlinear glossing with FLEx: conventions and challenges
- 4.9 Conversion to Universal Dependency (CoNLL-U) and gold-standard corpus creation
- 4.10 Corpus profile: size, genre, distribution of key markers

Chapter 5: Construction of the Morphosyntactic Filters and Prompts

- 5.1 Conceptualising the morphosyntactic filter
- 5.2 Rule extraction from descriptive sources and corpus evidence
- 5.3 Selection of the target constructions
- 5.4 Usage-based criteria: frequency, reliability, productivity
- 5.5 Functional criteria: information structure, argument structure, aspect
- 5.6 Filter architecture: analysis, constraint, and output stages
- 5.7 Prompt template design and Chain-of-Thought steps
- 5.8 Example prompts and walkthroughs
- 5.9 Anticipated effects on hallucination and projection

Chapter 6: Experimental Setup and Evaluation Protocol

- 6.1 Re-stating research questions in experimental terms
- 6.2 Models selected (GPT-4o, Claude 3.5 Sonnet, DeepSeek-V3, etc.)
- 6.3 Experimental conditions
 - 6.3.1 Baseline: zero-shot Universal Dependency parsing without linguistic rules
 - 6.3.2 Filtered: Universal Dependency parsing under morphosyntactic filter prompts
- 6.4 Data splits: development, test, and held-out sets
- 6.5 Evaluation metrics: UAS, LAS, and marker-specific accuracy
- 6.6 Implementation details and prompt execution
- 6.7 Ethical considerations, data security, and model usage constraints

Chapter 7: Results and Data Analysis

- 7.1 Overview of quantitative results across models and conditions
- 7.2 UAS/LAS comparisons: baseline vs filtered prompts
- 7.3 Performance on key morphosyntactic phenomena
- 7.4 Error typology in baseline outputs
- 7.5 Changes in error patterns under the morphosyntactic filter
- 7.6 Summary of analytical findings relative to research questions

Chapter 8: Findings and Discussions

13. Conclusion

Considering all this, this research aims to address the critical challenges in computational linguistics for low-resource languages- the systematic misanalysis of Santali in Large Language Models. Despite millions of speakers across Southeast Asia, Santali remains largely underrepresented in Large Language Models, and the models often hallucinate the outcomes due to the dominant languages' influence. The proposed morphosyntactic filter offers a linguistically grounded and theoretically encoded solution focusing on usage-based models, functional-typological grammar and constructional typology into carefully designed prompts. This approach harnesses the existing linguistic knowledge documented in grammars and corpora without requiring massive training datasets, making it scalable and pragmatic for low-resource languages. By incorporating linguistic theory and minority language perspectives, the proposed filter propounds a concrete pathway toward optimising advanced AI tools that respect the grammatical identity of vulnerable, endangered and under-resourced languages.

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